

R1

en

conf t

int gig 0/0

ip add 192.177.128.1 255.255.192.0

no shutdown

int se 0/0/0

ip add 192.178.80.1 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.177.128.0

network 192.178.80.0

en

conf t

int gig 0/0

ip helper-address 192.177.192.3

R2

```
en
conf t
int gig 0/0
ip add 192.178.64.1 255.255.248.0
no shutdown
int se 0/1/0
ip add 192.178.80.5 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.178.64.0
network 192.178.80.4
```

```
en
conf t
int gig 0/0
ip helper-address 192.177.192.3
```

R3

```
en
conf t
```

```
int gig 0/0
no ip address
no shutdown
int se 0/1/0
ip add 192.178.80.6 255.255.255.252
no shutdown
int se 0/0/0
ip add 192.178.80.2 255.255.255.252
no shutdown
int se 0/3/0
ip add 192.178.80.37 255.255.255.252
no shutdown
int se 0/2/0
ip add 192.178.80.41 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.178.72.0
network 192.178.76.0
network 192.178.78.0
network 192.178.80.0
network 192.178.80.4
network 192.178.80.36
network 192.178.80.40
```

switch

en

conf t

vlan 20

name CSE

exit

vlan 30

name Software

exit

vlan 40

name NFE

exit

show vlan brief

int range fa 0/1-5

switchport mode access

switchport access vlan 20

int range fa 0/6-12

switchport mode access

switchport access vlan 30

int range fa 0/13-18

```
switchport mode access
switchport access vlan 40
```

```
en
```

```
conf t
```

```
int fa0/24
```

```
switchport mode trunk
```

```
r
```

```
en
```

```
conf t
```

```
int gig0/0
```

```
no shut
```

```
int gig 0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.178.72.1 255.255.252.0
```

```
ip helper-address 192.177.192.3
```

```
int gig 0/0.30
```

```
encapsulation dot1Q 30
```

```
ip address 192.178.76.1 255.255.254.0
```

```
ip helper-address 192.177.192.3
```

```
int gig 0/0.40
```

```
encapsulation dot1Q 40
```

```
ip address 192.178.78.1 255.255.254.0
```

```
ip helper-address 192.177.192.3
```

R 4

```
en
```

```
conf t
```

```
int se 0/2/0
```

```
ip add 192.178.80.42 255.255.255.252
```

```
no shutdown
```

```
int se 0/0/0
```

```
ip add 192.178.80.17 255.255.255.252
```

```
no shutdown
```

```
int se 0/3/0
```

```
ip add 192.178.80.29 255.255.255.252
```

```
no shutdown
```

```
router rip
```

```
version 2
```

```
no auto-summary
```

```
network 192.178.80.16
```

```
network 192.178.80.28
```

```
network 192.178.80.40
```

R 5

en

conf t

int gig0/0

ip add 192.172.0.1 255.254.0.0

no shutdown

int se 0/3/0

ip add 192.178.80.30 255.255.255.252

no shutdown

int se 0/2/0

ip add 192.178.80.21 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.172.0.0

network 192.178.80.20

network 192.178.80.28

en

conf t

int gig 0/0

ip helper-address 192.177.192.3

R 6

en

conf t

int gig0/0

ip add 192.175.0.1 255.255.0.0

no shutdown

int se 0/2/0

ip add 192.178.80.22 255.255.255.252

no shutdown

int se 0/1/0

ip add 192.178.80.25 255.255.255.252

no shutdown

int se 0/0/0

ip add 192.178.80.77 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.175.0.0

network 192.178.80.20

network 192.178.80.24

network 192.178.80.76



```
en
conf t
int gig 0/0
ip helper-address 192.177.192.3
```

R 7

```
en
conf t
int se 0/1/0
ip add 192.178.80.26 255.255.255.252
no shutdown
int se 0/0/0
ip add 192.178.80.18 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.178.80.16
network 192.178.80.24
```

R 8

```
en
```

```
conf t
int gig0/0
ip add 192.176.0.1 255.255.0.0
no shutdown
int se 0/0/0
ip add 192.178.80.78 255.255.255.252
no shutdown
int se 0/1/0
ip add 192.178.80.73 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.176.0.0
network 192.178.80.72
network 192.178.80.76
```

```
en
conf t
int gig 0/0
ip helper-address 192.177.192.3
```

```
R 9
```

```
en
```

conf t

int se 0/1/0

ip add 192.178.80.74 255.255.255.252

no shutdown

int se 0/2/0

ip add 192.178.80.69 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.178.80.68

network 192.178.80.72

R 10

en

conf t

int gig0/0

ip add 192.177.0.1 255.255.128.0

no shutdown

int se 0/2/0

ip add 192.178.80.70 255.255.255.252

no shutdown

int se 0/1/0

ip add 192.178.80.65 255.255.255.252

no shutdown

int se 0/3/0

ip add 192.178.80.85 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.177.0.0

network 192.178.80.64

network 192.178.80.68

network 192.178.80.84

en

conf t

int gig 0/0

ip helper-address 192.177.192.3

R11

en

conf t

int se 0/3/0

ip add 192.178.80.38 255.255.255.252

no shutdown

int se 0/0/0

ip add 192.178.80.33 255.255.255.252

no shutdown

int se 0/2/0

ip add 192.178.80.13 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.178.80.36

network 192.178.80.12

network 192.178.80.32

R 12

en

conf t

int gig0/0

ip add 192.178.32.1 255.255.224.0

no shutdown

int se 0/0/0

ip add 192.178.80.34 255.255.255.252

no shutdown

int se 0/1/0

```
ip add 192.178.80.9 255.255.255.252
```

```
no shutdown
```

```
router rip
```

```
version 2
```

```
no auto-summary
```

```
network 192.178.32.0
```

```
network 192.178.80.8
```

```
network 192.178.80.32
```

```
en
```

```
conf t
```

```
int gig 0/0
```

```
ip helper-address 192.177.192.3
```

```
R 13
```

```
en
```

```
conf t
```

```
int gig0/0
```

```
ip add 192.174.0.1 255.255.0.0
```

```
no shutdown
```

```
int se 0/0/0
```

```
ip add 192.178.80.53 255.255.255.252
```

```
no shutdown
```

```
int se 0/2/0
ip add 192.178.80.14 255.255.255.252
no shutdown
int se 0/1/0
ip add 192.178.80.10 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.174.0.0
network 192.178.80.8
network 192.178.80.12
network 192.178.80.52
```

```
en
conf t
int gig 0/0
ip helper-address 192.177.192.3
```

R14

```
en
conf t
int se 0/0/0
```

```
ip add 192.178.80.54 255.255.255.252
```

```
no shutdown
```

```
int se 0/2/0
```

```
ip add 192.178.80.57 255.255.255.252
```

```
no shutdown
```

```
int se 0/1/0
```

```
ip add 192.178.80.49 255.255.255.252
```

```
no shutdown
```

```
router rip
```

```
version 2
```

```
no auto-summary
```

```
network 192.178.80.52
```

```
network 192.178.80.48
```

```
network 192.178.80.56
```

```
R15
```

```
en
```

```
conf t
```

```
int gig 0/0
```

```
ip add 192.168.1.1 255.254.0.0
```

```
no shutdown
```

```
int gig 0/1
```

```
ip add 192.170.0.1 255.254.0.0
```

```
no shutdown
```



```
int se 0/3/0
ip add 192.178.80.45 255.255.255.252
no shutdown
int se 0/1/0
ip add 192.178.80.50 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.168.1.0
network 192.170.0.0
network 192.178.80.44
network 192.178.80.48
```

```
en
conf t
int gig 0/0
ip helper-address 192.177.192.3
int gig 0/1
ip helper-address 192.177.192.3
```

R16

```
en
```

conf t

int se 0/2/0

ip add 192.178.80.58 255.255.255.252

no shutdown

int se 0/3/0

ip add 192.178.80.46 255.255.255.252

no shutdown

int se 0/0/0

ip add 192.178.80.61 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 192.178.80.56

network 192.178.80.44

network 192.178.80.60

R17

en

conf t

int se 0/0/0

ip add 192.178.80.62 255.255.255.252

no shutdown

int se 0/1/0

```
ip add 192.178.80.66 255.255.255.252
```

```
no shutdown
```

```
router rip
```

```
version 2
```

```
no auto-summary
```

```
network 192.178.80.60
```

```
network 192.178.80.64
```

```
R18
```

```
en
```

```
conf t
```

```
int se 0/3/0
```

```
ip add 192.178.80.86 255.255.255.252
```

```
no shutdown
```

```
int se 0/0/0
```

```
ip add 192.178.80.81 255.255.255.252
```

```
no shutdown
```

```
int se 0/1/0
```

```
ip add 192.178.80.89 255.255.255.252
```

```
no shutdown
```

```
router rip
```

```
version 2
```

```
no auto-summary
network 192.178.80.84
network 192.178.80.80
network 192.178.80.88
```

R19

```
en
conf t
int gig 0/0
ip add 192.177.224.1 255.255.224.0
no shutdown
int se 0/0/0
ip add 192.178.80.82 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.177.224.0
network 192.178.80.80
```

```
conf t
int gig 0/0
ip helper-address 192.177.192.3
```

en

conf t

access-list 1 permit 192.168.0.0 0.1.255.255

access-list 1 permit 192.170.0.0 0.1.255.255

access-list 1 permit 192.172.0.0 0.1.255.255

access-list 1 permit 192.174.0.0 0.0.255.255

access-list 1 permit 192.175.0.0 0.0.255.255

access-list 1 permit 192.176.0.0 0.0.255.255

access-list 1 permit 192.177.0.0 0.255.255.255

access-list 1 permit 192.178.0.0 0.255.255.255

interface s0/0/0

ip nat inside

exit

interface g0/0

ip nat outside

exit

ip nat inside source list 1 interface g0/0 overload

R20

en

conf t

```
int gig 0/0
ip add 192.178.0.1 255.255.224.0
no shutdown
int se 0/0/0
ip add 192.178.80.93 255.255.255.252
no shutdown
int se 0/1/0
ip add 192.178.80.90 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 192.178.0.0
network 192.178.80.88
network 192.178.80.92
```

```
conf t
int gig 0/0
ip helper-address 192.177.192.3
```

R21

en

```
conf t
```

```
int gig 0/0
```

```
ip add 192.177.192.1 255.255.224.0
```

```
no shutdown
```

```
int se 0/0/0
```

```
ip add 192.178.80.94 255.255.255.252
```

```
no shutdown
```

```
router rip
```

```
version 2
```

```
no auto-summary
```

```
network 192.177.192.0
```

```
network 192.178.80.92
```

```
conf t
```

```
int gig 0/0
```

```
ip helper-address 192.177.192.3
```

```
http://www.instagram.com/instagram.html
```

```
http://www.facebook.com/facebook.html
```

```
http://www.whatsapp.com/whatsapp.html
```

```
http://www.youtube.com/youtube.html
```